adesso.it

From «Hello, world!» to distributed systems

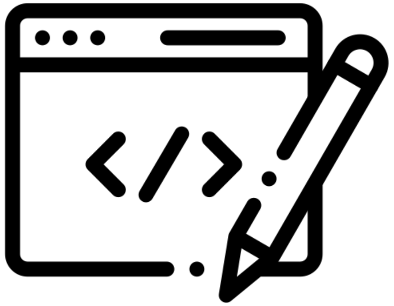
A practical overview of the key points to master if you want to be a good dev!

Jacopo Di Stanislao
Software Architect @ adesso.it

Agenda

1. A software developer journey
2. Well coded software
3. Demo
4. Well designed software
5. Demo
6. Q&A

A software developer journey

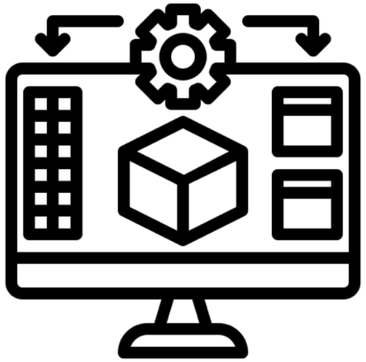


Junior
Developer

Code design

Senior
Developer

Software design



Software
Architect

System design

Well coded software

- Clear
- Evolvable
- Correct
- Simple

Well coded software

// **Clear**

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- O(n)

// Evolvable

// Correct

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- O(n)

// **Evolvable**

- Encapsulation
- Cohesion vs Coupling
- SOLID
- Design Patterns

// Correct

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- O(n)

// Evolvable

- Encapsulation
- Cohesion vs Coupling
- SOLID
- Design Patterns

// **Correct**

- Unit testing
- Integration testing
- Code reviews
- TDD

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- O(n)

// Evolvable

- Encapsulation
- Cohesion vs Coupling
- SOLID
- Design Patterns

// Correct

- Unit testing
- Integration testing
- Code reviews
- TDD

// **Simple**

- DRY
- KISS
- Fits in your head
- Good enough

Well coded software

> "Talk is cheap, show me the code!"
>
> *(Linus Torvalds)*

Coding inside a team

// Version Control

- Local vs. remote repository
- Branches
- Fork and merge
- Branching strategies
- Repo history
- Tags

Well designed software

- Deployment Model
- Sync vs. Async
- Distributed Systems
- Observability

Well designed software

// **Deployment Model**

- Monolith
- Microservices

// Sync vs. Async

// Distributed Systems

// Observability

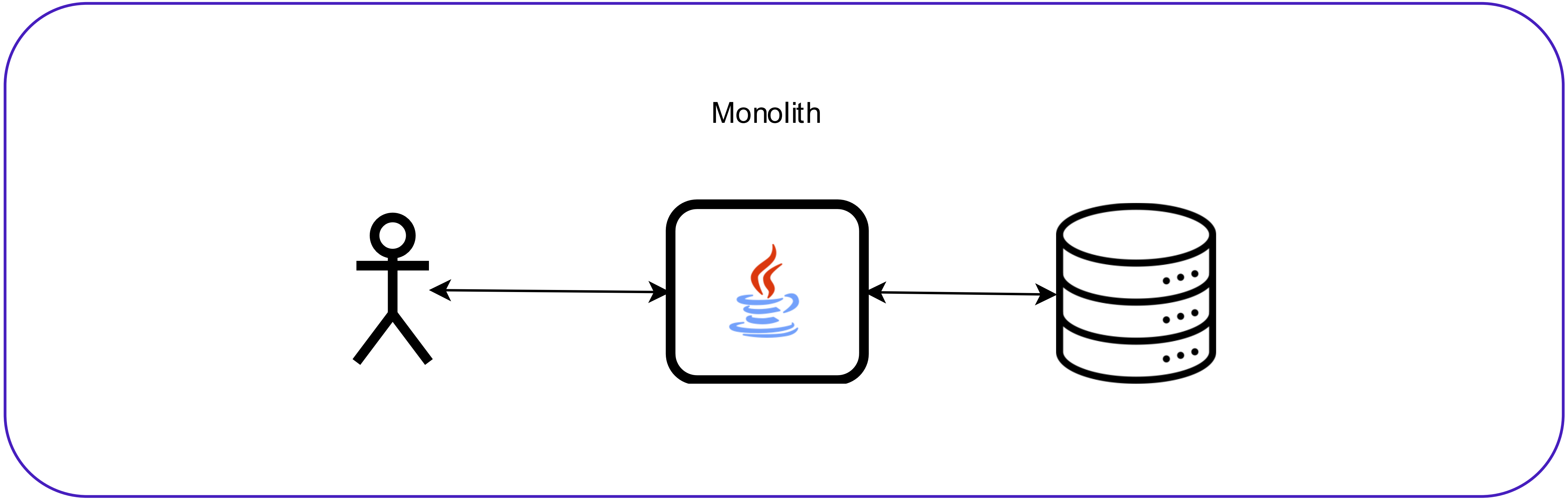
Well designed software

// **Deployment Model**

// Sync vs. Async

// Distributed Systems

// Observability



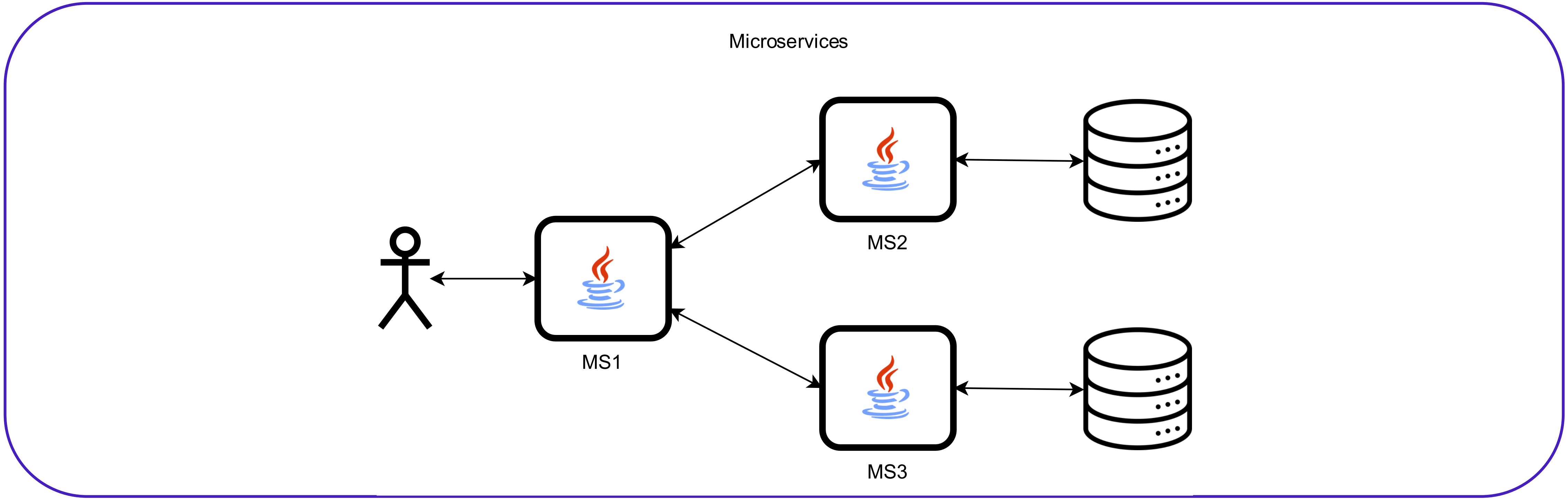
Well designed software

// **Deployment Model**

// Sync vs. Async

// Distributed Systems

// Observability



Well designed software

// Deployment Model

- Monolith
- Microservices

// **Sync vs. Async**

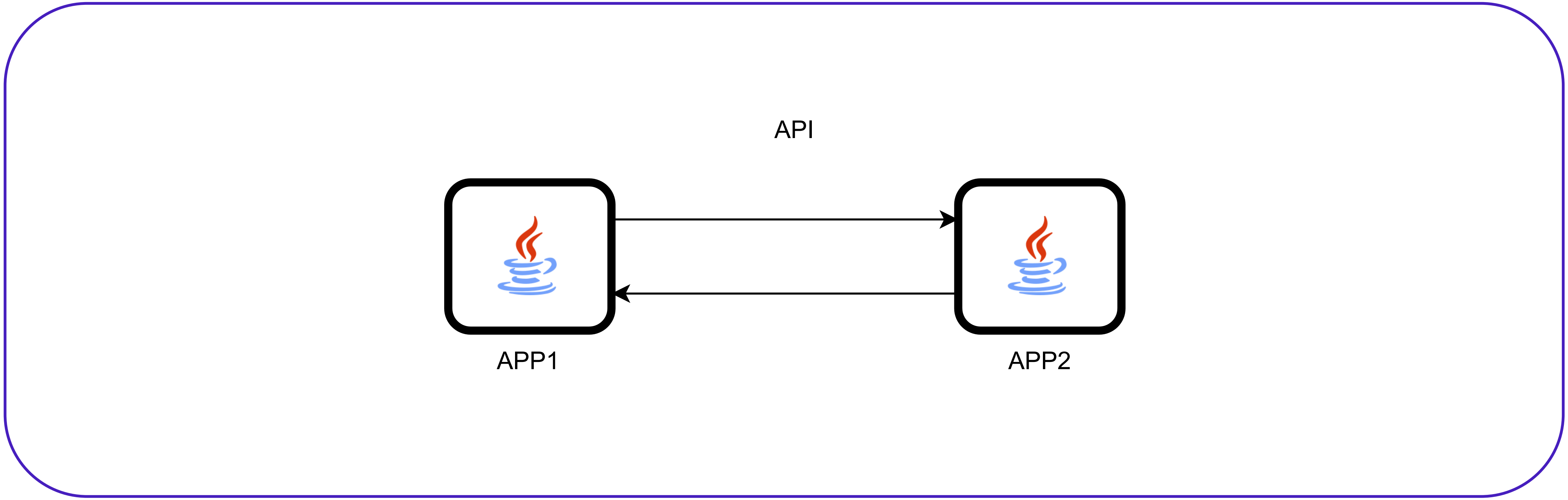
- API
- Messages
- Idempotency

// Distributed Systems

// Observability

Well designed software

- Deployment Model
- **Sync vs. Async**
- Distributed Systems
- Observability



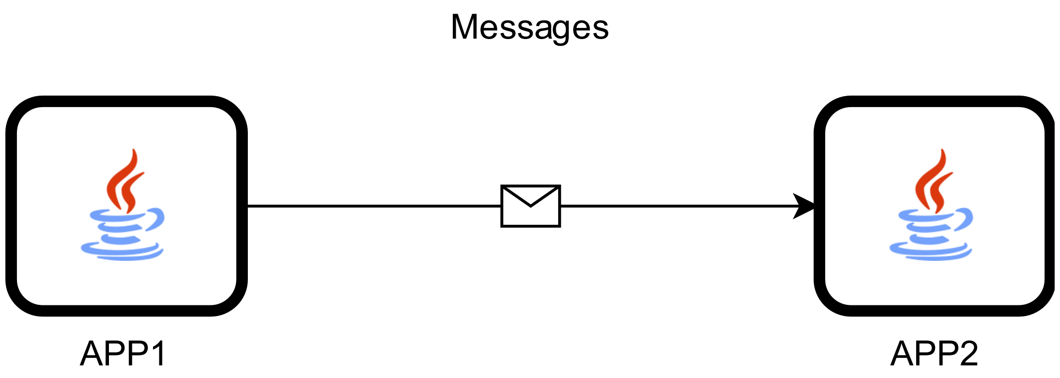
Well designed software

// Deployment Model

// **Sync vs. Async**

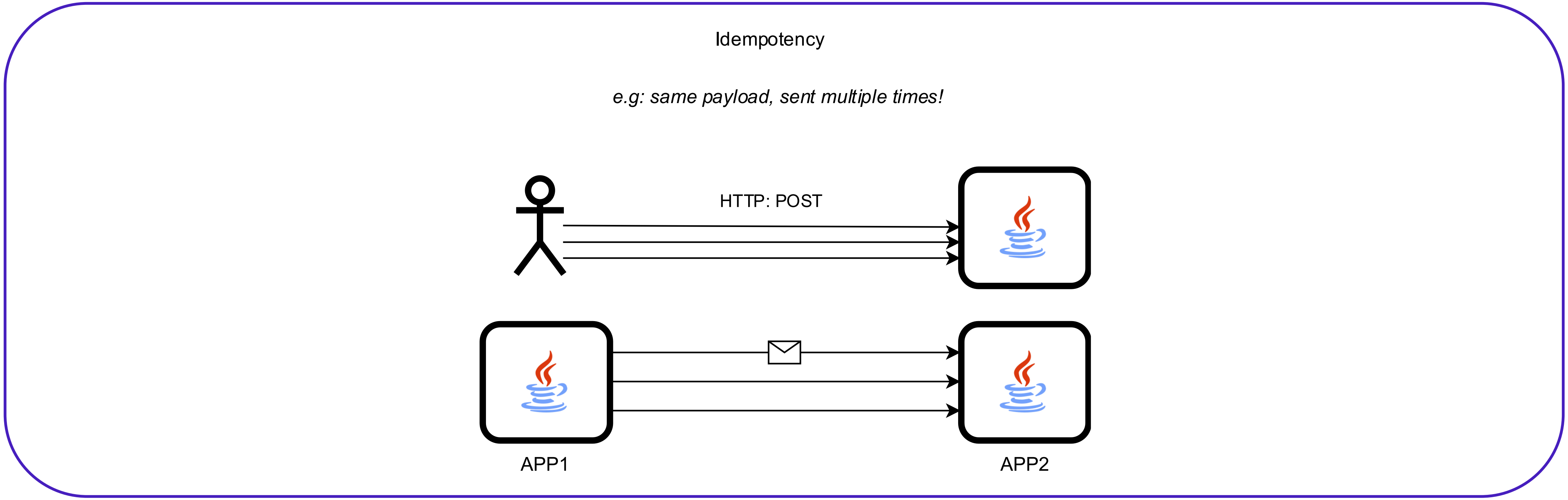
// Distributed Systems

// Observability



Well designed software

- // Deployment Model
- // **Sync vs. Async**
- // Distributed Systems
- // Observability



Well designed software

// Deployment Model

- Monolith
- Microservices

// Sync vs. Async

- API
- Messages
- Idempotency

// **Distributed Systems**

- Transactions
- Errors
- Eventual consistency

// Observability

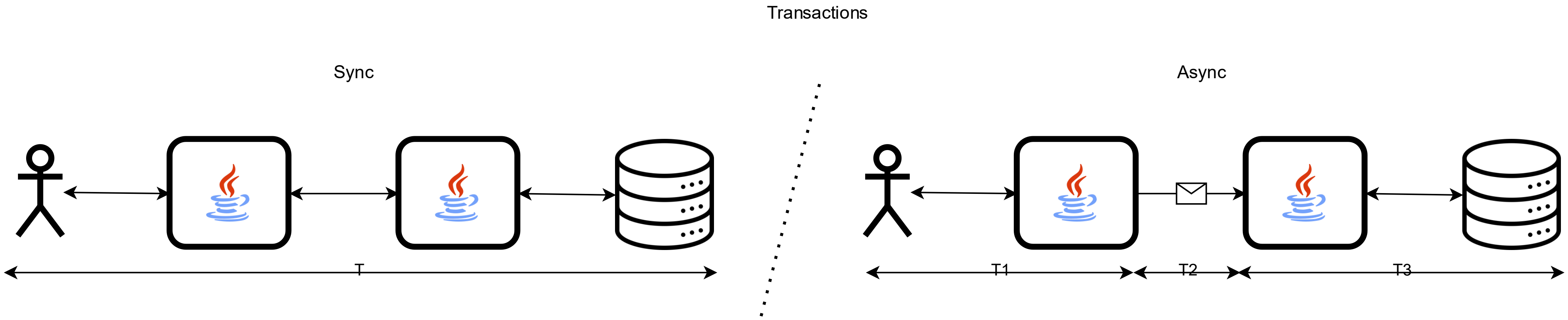
Well designed software

// Deployment Model

// Sync vs. Async

// **Distributed Systems**

// Observability



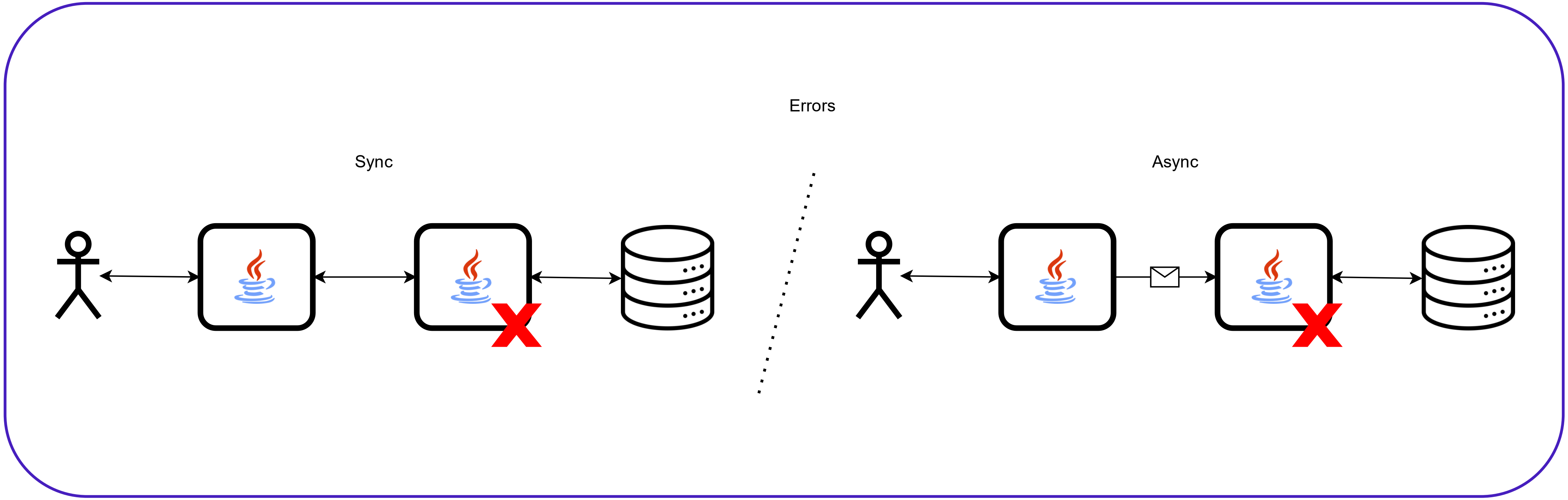
Well designed software

// Deployment Model

// Sync vs. Async

// **Distributed Systems**

// Observability



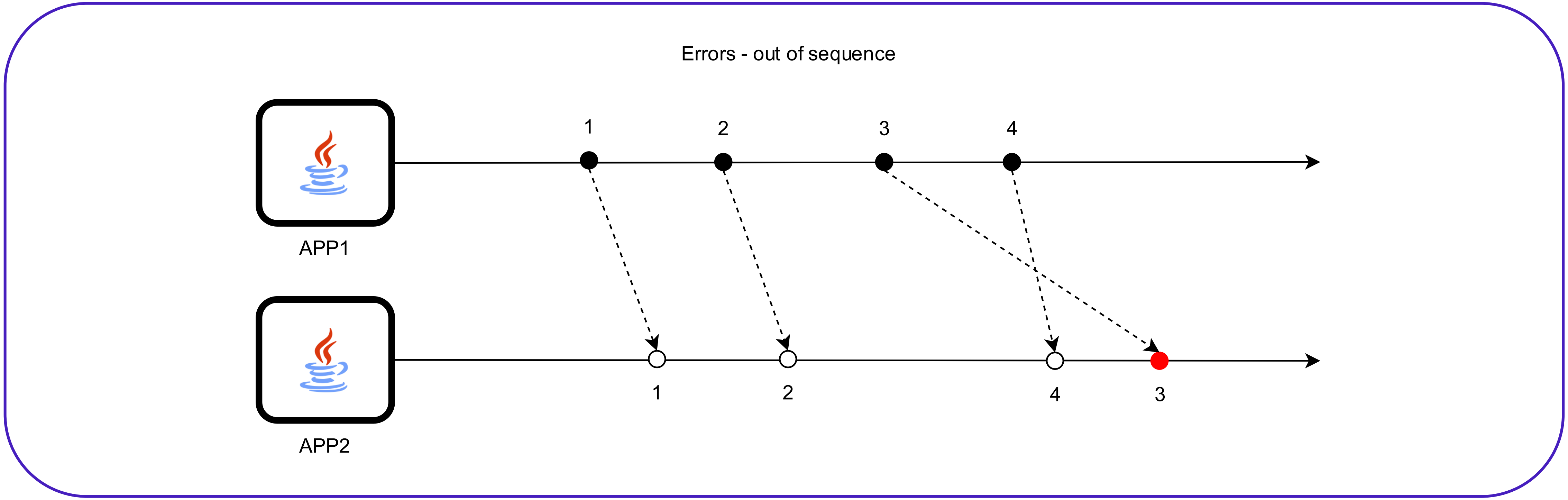
Well designed software

// Deployment Model

// Sync vs. Async

// **Distributed Systems**

// Observability



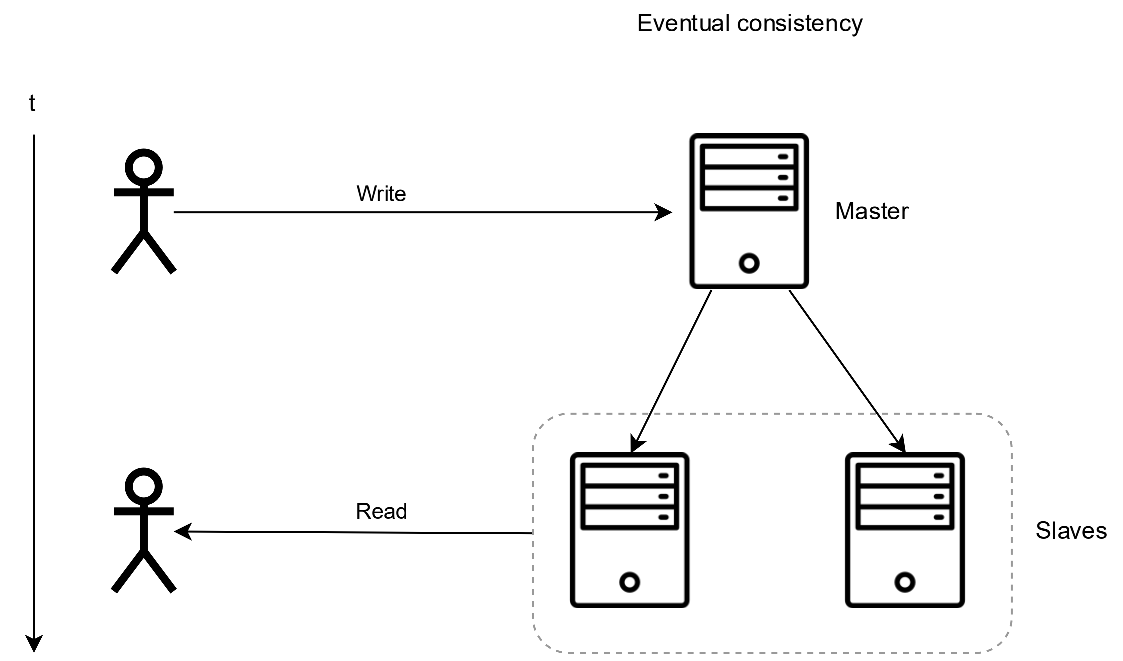
Well designed software

// Deployment Model

// Sync vs. Async

// **Distributed Systems**

// Observability



Well designed software

// Deployment Model

- Monolith
- Microservices

// Sync vs. Async

- API
- Messages
- Idempotency

// Distributed Systems

- Errors
- Transactions
- Eventual consistency

// **Observability**

- Logs
- Metrics
- Tracing

Well designed software

“Demo!”

Q&A

adesso.it

Grazie mille.

www.linkedin.com/company/adesso-it

www.adesso.it

info@adesso.it

MILANO

Viale Edoardo Jenner, 51 20159
Milano, Italia

+39 02 697731

LECCE

Via Merine, 36 73100
Lecce, Italia

+39 0832 1778061

adesso.it